

GILLAM, MB, Canada

WMO: 719120

Lat: 56.360N

Lon: 94.710W

Elev: 476

StdP: 14.44

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.4	-30.9	-42.2	0.5	-34.1	-38.8	0.6	-30.5	24.4	-3.7	21.1	-4.4	7.8	270	0.790

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	81.2	64.0	77.8	62.4	74.4	61.1	67.1	76.5	64.8	73.7	62.9	71.6	10.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	90.2	71.0	61.3	82.4	68.8	59.2	76.3	67.2	32.0	76.5	30.2	73.6	28.7	71.9	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	21.0	18.7	-39.9	89.1	4.0	3.7	-42.8	91.7	-45.1	93.8	-47.4	95.9	-50.3	98.5
			-40.1	72.2	3.9	3.5	-42.9	74.7	-45.2	76.7	-47.4	78.7	-50.3	81.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.4	-10.8	-6.5	6.2	24.0	39.0	54.0	61.2	58.3	47.9	33.3	12.3	-4.1
	DBStd	27.58	13.08	12.44	14.25	13.34	10.37	8.68	6.52	6.54	7.94	8.00	12.63	14.28
	HDD50	9503	1886	1583	1356	782	368	56	3	7	134	520	1132	1676
	HDD65	14154	2351	2003	1821	1230	806	344	152	226	515	983	1582	2141
	CDD50	893	0	0	0	2	28	175	351	265	70	2	0	0
	CDD65	70	0	0	0	0	1	13	35	19	2	0	0	0
	CDH74	819	0	0	0	0	22	198	378	196	24	1	0	0
	CDH80	159	0	0	0	0	1	39	75	40	4	0	0	0
	WSAvg	9.1	8.6	8.8	8.7	9.0	9.5	8.9	8.7	9.3	10.0	10.4	9.4	8.4
	PrecAvg	20.4	0.8	0.7	0.9	1.2	1.9	2.2	3.4	3.0	2.2	1.6	1.4	1.0
(15) Precipitation	PrecMax	28.3	1.5	1.4	1.6	2.8	5.4	5.0	6.4	5.0	3.7	4.1	3.2	1.8
	PrecMin	13.9	0.2	0.2	0.2	0.2	0.2	0.6	1.2	0.7	1.3	0.3	0.5	0.3
	PrecStd	4.3	0.4	0.3	0.4	0.6	1.2	1.1	1.5	1.3	0.8	0.9	0.8	0.5
	PrecStd	4.3	0.4	0.3	0.4	0.6	1.2	1.1	1.5	1.3	0.8	0.9	0.8	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.4	33.3	44.8	64.5	77.5	84.7	87.1	85.0	77.2	59.8	40.2	30.5
		MCWB	27.5	31.5	37.7	49.6	58.4	65.2	69.2	67.3	62.9	49.3	36.7	29.5
	2%	DB	21.2	25.7	39.1	54.3	69.4	79.4	81.5	79.0	70.8	53.2	35.3	25.4
		MCWB	20.3	23.8	33.9	43.0	54.0	61.2	64.4	64.4	59.8	47.2	33.3	24.5
	5%	DB	15.7	19.1	33.9	48.4	62.8	75.3	78.1	75.1	65.4	48.3	32.4	21.3
		MCWB	15.0	17.6	30.9	39.7	50.6	59.4	63.1	62.8	56.8	44.2	31.3	20.5
	10%	DB	9.6	13.1	28.7	43.0	57.0	70.9	74.8	70.9	60.5	44.2	29.6	17.7
		MCWB	8.9	11.9	26.1	36.3	47.2	57.7	62.0	60.7	54.1	41.2	28.6	16.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.5	31.7	38.1	50.3	59.4	70.3	71.2	69.4	65.0	52.0	37.3	29.5
		MCDB	28.3	33.4	43.7	63.9	72.4	79.8	82.9	79.7	73.2	57.4	39.4	30.2
	2%	WB	20.4	23.8	34.3	44.1	55.2	64.4	67.3	66.6	61.1	47.8	33.7	24.6
		MCDB	21.2	25.7	38.4	52.9	67.4	74.3	76.7	75.6	69.3	52.6	35.0	25.4
	5%	WB	15.0	17.5	30.9	40.1	51.4	61.4	65.1	64.3	57.6	44.5	31.4	20.5
		MCDB	15.7	19.0	34.0	47.6	61.1	72.0	74.1	72.5	64.2	48.0	32.3	21.2
	10%	WB	8.9	11.9	25.9	36.5	47.8	58.7	63.2	62.1	54.8	41.4	28.7	17.0
		MCDB	9.6	13.2	28.6	42.4	56.4	68.9	72.3	69.2	60.1	43.9	29.7	17.7
(35) Mean Daily Temperature Range	5% DB	MDBR	14.7	18.8	22.0	21.7	20.1	21.9	20.2	17.9	15.6	10.8	12.6	13.7
		MCDBR	16.4	21.9	22.9	26.0	30.6	28.3	25.7	24.8	24.2	18.8	11.5	14.4
		MCWBR	16.1	20.3	18.5	17.0	17.7	13.9	12.5	13.2	14.2	13.4	10.5	14.2
		MCDBR	16.5	21.2	21.2	25.0	28.0	24.6	21.8	21.4	21.2	17.5	11.3	14.3
	5% WB	MCWBR	16.3	19.7	17.4	16.8	17.2	14.0	12.0	12.8	13.8	12.9	10.5	14.1
(40) Clear-Sky Solar Irradiance	taub		0.230	0.262	0.271	0.333	0.340	0.354	0.403	0.358	0.326	0.285	0.259	0.212
	taud		2.208	2.218	2.271	2.221	2.400	2.407	2.322	2.442	2.514	2.545	2.246	2.159
	Ebn at Noon		221	257	286	280	285	282	264	270	265	252	203	196
	Edh at Noon		16	25	30	38	34	34	37	30	25	18	16	13
(44) All-Sky Solar Radiation	RadAvg		256	605	1126	1669	1763	1846	1714	1343	895	456	220	162
	RadStd		20	44	88	140	146	179	111	125	86	60	22	20

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page